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# **Secure OPC UA Pub/Sub**

## DSD project proposal

# ABB Process Automation (PA)

## Process Control Platform (PCP)

- Size: ABB ~105 000, PA ~ 21 000, PCP ~1 000 (~300 in Sweden)
- PCP - R&D department of Process Automation
- Developing products and systems for Distributed Control Systems
  - Operation workplace
  - Engineering tools
  - Controllers
  - Fieldbus connectivities & network infrastructure
  - I/O
- Software, Firmware, Hardware, Electronics, Mechanics
- Process & Hybrid industries
  - Power Generation, Mining, Oil & Gas, Marine, Chemicals
- My role: Cybersecurity Architect – Control I/O Networks



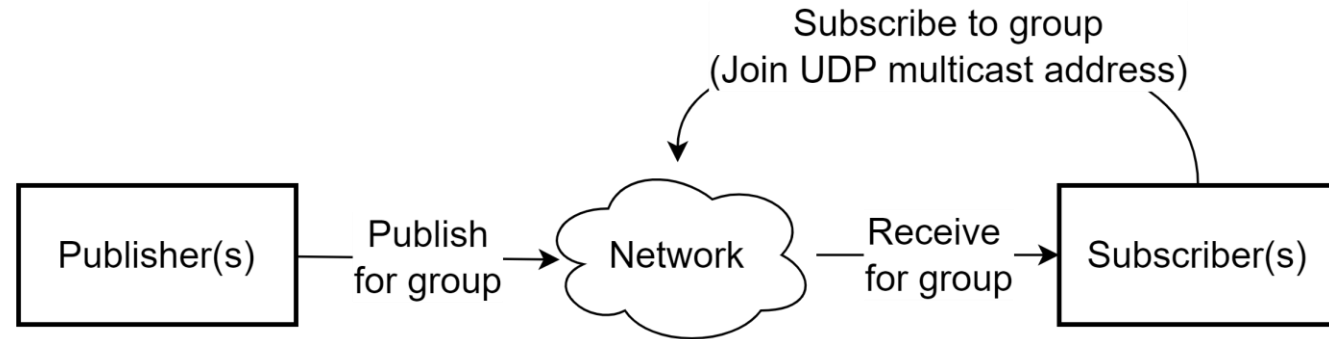
# About OPC UA pub/sub

Open industrial protocol

Publish / subscribe pattern

Message oriented middleware

- Network as middleware
- UDP Multicast as mechanism



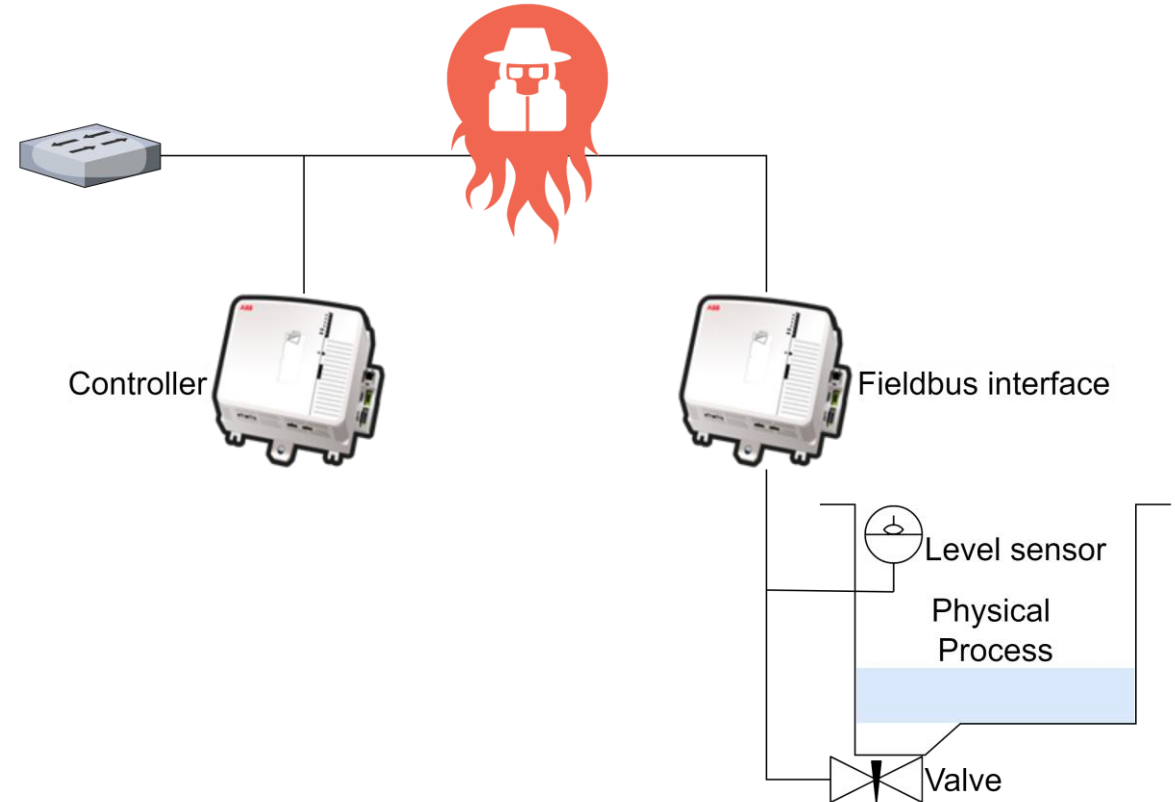
# Why is OPC UA pub/sub important?

Horizontal data exchange

- Communicate signals for actuators/sensors
- Cross-Controller synchronizations
- Heartbeat for redundancy protocols

Problem: Security – limited support in stacks & testbenches

- Non-secure pub/sub over UDP multicast:
  - No integrity
  - No confidentiality
  - Attacks – potential safety impact



# About the project

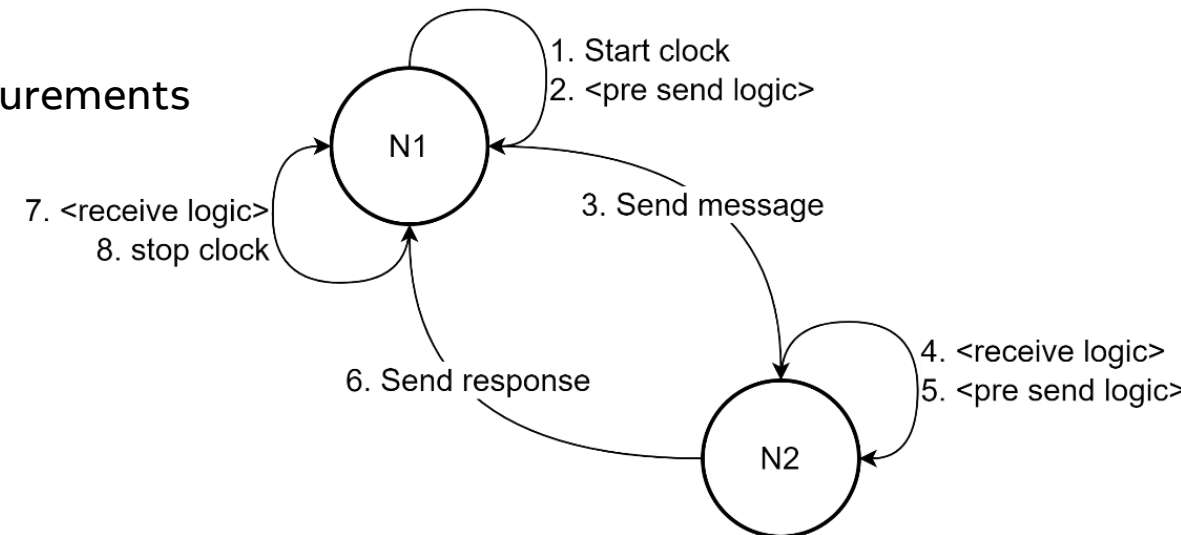
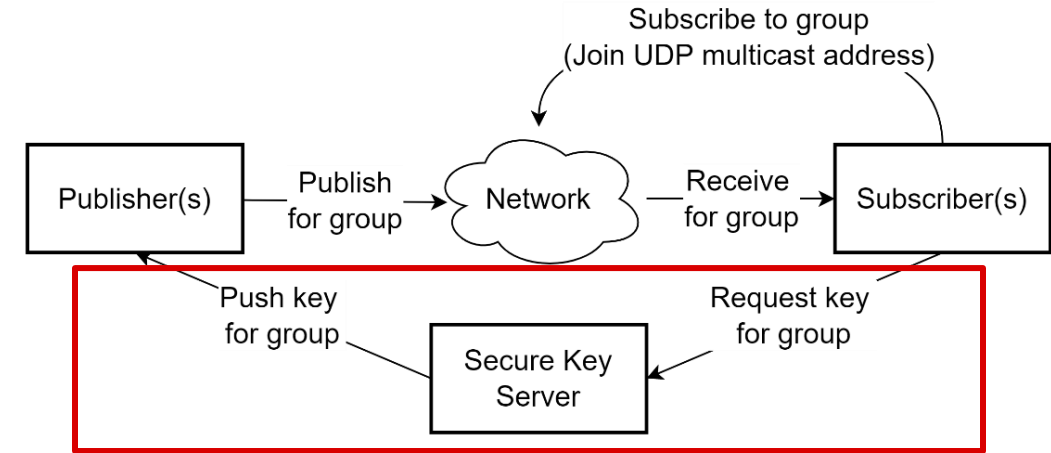
Extend an existing prototype implementation

**Goal: Understand if secure pub/sub can be used the way we want**

- Productify prototype
  - Configuration handling in general – now everything is hardcoded
  - Certificates
- Extend Key Server
  - DB for local key storage
  - User/service authentication and authorization
- Extend Publisher/Subscriber to support round-trip time measurements
  - Prepare / execute initial experiments
- Variable deployment scenarios

## Environment:

- C++, Google test, GitHub
- Based on Open62541 project
- Linux/WSL



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# Thank you for listening!



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**ABB**